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The green potential of artificial intelligence: revisiting energy consumption, growth, and ecological footprint in the United States

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Despite AI's rapid progress of AI as a tool for equitable growth, little is known about how it can help to slow environmental damage. From 1996 to 2024, this study examined how the US's ecological footprint is impacted by innovations in AI, energy use, economic development, industrialization, and growing populations. After confirming the mixed order of integration, the study applies the Autoregressive Distributed Lag (ARDL) method to evaluate short-term and long-term trends. Robustness was tested using three techniques: Canonical Cointegration Regression, Dynamic Ordinary Least Squares, and Fully Modified Ordinary Least Squares. The results show that the ecological footprint is positively affected by economic growth, increased energy consumption, industrialization, and population growth, all of which underline the environmental impacts of economic and demographic advancement. In contrast, AI innovation reduces ecological pressure, demonstrating its potential to optimize energy efficiency, encourage cleaner production, and enhance overall environmental quality. These results are consistent across both the short- and long-term calculations and robustness tests. This study contributes to the growing debate on technology and sustainability by positioning artificial intelligence as the key driver of ecological resilience. Policy implications stress the urgency of accelerating AI-driven green strategies, investing in renewable energy, and fostering sustainable industrial practices to balance economic progress with environmental preservation in the United States.

KEYWORDS
ARDL, artificial intelligence, energy, environmental quality, United States

1 Introduction

Over the last few decades, global warming, rising worldwide temperatures, and increasing ecological constraints have become pressing challenges. Economic prosperity is usually pursued at the cost of increasing energy use, and it is well known that energy use is one of the key contributors to biodiversity loss (Joof et al., 2024). The United States plays a very important role in this discourse. As the second-largest carbon dioxide emitter, it accounts for approximately 16% of global energy, but it accounts for 4.3% of the global population (Husain et al., 2024). This imbalance results in an unbalanced environmental load in relation to the country's economic magnitude. New statistics indicate improvements and ongoing challenges. In 2023, U.S. energy-related CO_2 emissions stood at 4.8 billion metric tons, which is 2.7% lower than the previous year and almost 20% less than in 2005 (Tiseo, 2024). These improvements largely reflect advances in technology and growing investment in renewable energy capacity, which has expanded by 13% worldwide by 2022 (IEA, 2023). Simultaneously, fossil fuels still dominate the energy mix in the U.S., as they still cause upward pressure on ecological stress. Against this background, the ecological footprint (EF) presents a more holistic metric of ecosystem conditions than carbon emissions alone, as it considers the resource breadth of economic development, energy consumption, and industrialization (Dogan et al., 2024; Javeed et al., 2023). This perspective underscores the need to explore new drivers of sustainability in the U.S. context, particularly those that extend beyond conventional economic and demographic factors.

There has been extensive research on the relationship between the environment and the process of economic activity degradation. There is a vast body of literature on how economic growth, energy use, industrialization, and urbanization affect the health of ecosystems, frequently using CO_2 emissions as an important indicator of ecological stress (Ahmad et al., 2024; Akther et al., 2025). While these studies have generated important insights, the narrow use of CO_2 emissions risks overlooks the thorough evaluation of sustainability, as it reflects the aggregate demand on natural resources beyond emissions alone (Zafar et al., 2019; Nketiah et al., 2024). Another limitation lies in the underrepresentation of technological innovation in environmental research. Although economic and demographic factors have been well studied, few studies have systematically examined artificial intelligence as a determinant of ecological sustainability (Raihan et al., 2024b). However, AI has the potential to transform energy systems, industrial efficiency, and environmental monitoring (Akhter et al., 2024; Ridwan et al., 2024a). The available literature that incorporates technology into environmental models is still disparate by region and methodology, with significant gaps in the knowledge of the distinct role of AI in high-income, high-emission economies such as the United States.

Artificial intelligence (AI) has become a driving technological issue with wide implementation in the management of energy, industrial optimization, and environmental monitoring. AI has the potential to make the production process energy-efficient, waste-reductive, and facilitated by automation, contributing to the shift to a low-carbon production system, which is possible by making it predictive analytics, enabling the distribution of resources in real time, and enhancing the automation level (Liu et al., 2024). In addition to industrial applications, AI is useful for conserving biodiversity, monitoring the environment, and managing climate risks by enhancing tracking and surveillance, and making decisions based on these data (Su et al., 2021; Ridwan et al., 2024b). Such abilities make AI not only an economic force but also a potential solution to ecological degradation. Despite this potential, the use of AI in the development of ecological sustainability has had minimal empirical coverage in the literature. An increasing, yet limited, body of literature demonstrates that AI innovation contributes to environmental quality in various regional settings. An example is a study in the United States, G-7, and Nordic countries, which invariably reports that the adoption of AI leads to better ecological results in the short and long term (Ridwan et al., 2024c; Zafar et al., 2019). Other evidence from China and Europe indicates that the integration of AI will allow the decoupling of growth and environmental degradation (Zhao et al., 2023). Nevertheless, these investigations are still sporadic, as most tend to examine smaller environmental variables or a smaller geographic area, and the critical issue of whether AI can help alleviate ecological pressures in economies with high emissions, including the United States, remains unanswered.

To address these gaps, this study examines how the EF of the United States in the 1996–2024 periods depends on AI innovation, energy usage, growing GDP, industrialization, and population growth. This study offers a more global view of the implications of financial and technological developments in the environment by placing emphasis on the ecological footprint rather than on CO_2 emissions. The analysis is methodologically sound and uses an efficient econometric framework. A number of unit root tests were conducted to define the sequence of integration and to estimate both short- and long-term relationships; additionally, ARDL, FMOLS, DOLS, and CCR were performed as further estimates to enhance the findings. This research has two main contributions to the literature: First, it integrates artificial intelligence into the environmental–economic nexus, offering fresh evidence of its potential to reduce biodiversity loss in a high-income, high-emission economy. Second, it provides policy-relevant insights by identifying how technological innovation can complement renewable energy expansion and sustainable industrial practices to reduce ecological pressures. By doing so, this study advances the understanding of technology-driven sustainability pathways in the United States.

This research offers several important contributions to the scholarly work and policy debates on sustainability. Theoretically, it builds on the current body of research on the causes of environmental degradation by directly considering artificial intelligence as a technological factor that determines the ecological footprint. Although other researchers have focused on more traditional indicators, such as energy consumption, GDP, and population, this study focuses on the duality of AI as an economic facilitator and an ecological shield. Methodologically, this study employs an advanced econometric methodology that integrates the ARDL model with various robustness test analyses to provide credible short- and long-run estimates. This method will assist in developing a better understanding of the temporal dynamics of ecosystem factors in the United States, a nation that remains dominant in the economic and energy circles worldwide. This study has practical implications for policy. The findings are important, as they prove the necessity of integrating AI-based innovation into U.S. sustainability strategies by showing that AI can alleviate ecological pressures. Such policies may include investing in smart energy management systems and encouraging AI-driven industrial efficiency and eco-innovation with specific incentives. Beyond the context of the U.S., the results can also be applied to other economies with high incomes and high emissions that are grappling with the balance between economic growth and environmental protection.

The remainder of this paper is organized as follows. Section 2 covers the available literature on the selected variables. Section 3 explains the data and the methodological framework used in the analysis. Section 4 provides the empirical findings and discussions with strong checks and diagnostics. Finally, Section 5 presents the implications of the study and Section 6 summarizes the research by pointing out its main findings, policy recommendations, and future research directions.

2 Literature review and hypothesis development

Understanding the drivers of the ecological footprint requires a critical review of existing studies that link economic, demographic, and technological factors with environmental outcomes. Past studies on economic growth, energy consumption, industrialization, and population formation have been identified as major contributors to ecological degradation, whereas recent studies also emphasize the importance of artificial intelligence in determining the direction of sustainability. Building on this literature, the following subsections review prior findings and formulate testable hypotheses that guide the empirical analysis of the United States case.

2.1 Economic growth and ecological footprint

The relationship between economic growth and environmental sustainability has long been debated in the literature. A substantial body of empirical evidence suggests that rising GDP levels tend to intensify environmental pressure through increased energy demand, expanded industrial production, and greater resource extraction. Rahman and Alam (2021) found that higher GDP levels in Bangladesh significantly increased ecological harm, while Ahmad et al. (2024) reported similar results for China, where economic growth contributed to greater carbon emissions. Raihan et al. (2024b) further demonstrated that GDP expansion in G-7 economies heightened environmental pressure. Comparable findings have also been documented for the United States, China, Belt and Road Initiative economies, and BRICS countries, where growth was associated with rising environmental degradation (Joof et al., 2024; Su et al., 2021; Ali et al., 2022; Saud et al., 2019). Nevertheless, the growth and environment nexus is not universally linear or uniformly negative. Evidence from G seven countries suggests that economic expansion may under certain conditions contribute to ecological improvement, particularly when accompanied by structural transformation toward service sectors, technological upgrading, and cleaner energy adoption (Mehmood et al., 2023; Adebayo, 2024). This divergence reflects the context specific nature of the growth and environment relationship, especially in advanced economies such as the United States, where high technological intensity and structural change may alter traditional scale effects. Accordingly, the long run association between GDP and ecological footprint remains an empirical question in the U.S. setting, warranting a country level investigation within a unified framework.

H1: Economic growth increases the ecological footprint in the United States.

2.2 Energy use and ecological footprint

Energy consumption is widely regarded as a critical determinant of ecological degradation. Heavy reliance on fossil fuels contributes directly to carbon emissions and places mounting pressure on ecological systems. Nathaniel et al. (2021) demonstrated that increased energy use amplifies ecological footprint in Next Eleven economies, while Raihan et al. (2024a) found that rising energy consumption in Vietnam deteriorates environmental quality. Similar evidence from the United Kingdom confirms that higher energy use exerts a significant and positive effect on ecological footprint (Eweade et al., 2024). In Italy, Pattak et al. (2023) observed that a one percent increase in non-renewable energy consumption corresponds to a 1.505 percent rise in carbon emissions, further highlighting the environmental burden associated with fossil based energy structures. However, the energy and environment nexus is not homogeneous across energy types. Rahman et al. (2023) showed that sustainable energy adoption substantially reduces ecological footprint in populous countries, while Sun et al. (2023) revealed heterogeneous outcomes across BRICS nations, where renewable energy sources offset environmental damage in some contexts but not universally. These findings suggest that the environmental implications of energy consumption depend not only on quantity but also on composition and structural transition. In the United States, where fossil fuels continue to account for a significant share of industrial and household energy demand, the relationship between energy use and ecological footprint remains particularly relevant (Fatima et al., 2024). Although renewable energy expansion has accelerated in recent years, the persistence of carbon intensive infrastructure may sustain ecological pressure. This mixed evidence underscores the importance of re-examining the long run association between total energy consumption and ecological footprint within the U.S. context.

H2: Energy consumption increases the ecological footprint in the United States.

2.3 Industrialization and ecological footprint

Industrialization has played a central role in economic and financial development, yet its environmental consequences remain deeply contested. The expansion of industrial activities increases energy demand, accelerates fossil fuel combustion, and contributes directly to greenhouse gas emissions. Empirical evidence across regions confirms that industrialization often intensifies ecological pressure. Usman and Hammar (2021) found that industrialization significantly deteriorated environmental quality in Asia Pacific Economic Cooperation economies over the long run, while Aslam et al. (2024) reported that industrial expansion in East Asian and Pacific states substantially increased ecological footprint. Similar outcomes have been documented in BRICS, OECD, and MENA countries, where industrialization has been associated with ecosystem decline and rising environmental stress (Kongbuamai et al., 2021; Nasrollahi et al., 2020). However, the industrialization and environment nexus is not universally uniform.

Patel and Mehta (2023) showed that industrialization in India exhibited dynamic effects over time, suggesting that structural reforms and the adoption of cleaner technologies can alter the traditional pollution intensive trajectory of industrial growth. These findings imply that the environmental impact of industrialization depends on technological composition, regulatory frameworks, and the degree of energy transition embedded within industrial structures. In the United States, industrial activity remains technologically advanced yet energy intensive in several sectors. Although modernization and regulatory standards may reduce pollution intensity, the scale and organization of industrial production can continue to exert ecological pressure. This complexity highlights that the long run relationship between industrialization and ecological footprint in the U.S. context is not predetermined but contingent upon structural and technological conditions. Accordingly, examining industrialization within a unified empirical framework is essential for understanding its aggregate sustainability implications.

H3: Industrialization increases the ecological footprint in the United States.

2.4 Population and ecological footprint

Population growth is widely regarded as a fundamental driver of ecological degradation, as it intensifies demand for food, housing, transportation, and energy. A larger population increases direct consumption requirements while simultaneously stimulating industrial production and urban expansion, thereby amplifying environmental stress. Empirical evidence strongly supports this scale effect. Khan et al. (2023) found that demographic expansion significantly deteriorated environmental quality in South Asian economies, while Rahman and Alam (2021) reported similar outcomes for Bangladesh. Evidence from African countries also indicates that population growth expands ecological footprint and carbon emissions through rising resource demand (Nathaniel et al., 2021). Nevertheless, the population and environment relationship is not mechanically proportional. Dogan et al. (2019) observed that in certain advanced economies, technological progress and resource efficiency improvements can partially offset the ecological consequences of demographic expansion. Similarly, Sun et al. (2023) identified heterogeneous outcomes across BRICS nations, where the environmental impact of population growth depended on energy structure and governance quality. These findings suggest that demographic pressure interacts with institutional capacity, technological intensity, and energy composition rather than operating in isolation. In the United States, population growth occurs alongside high *per capita* income and consumption levels, which may magnify ecological demand relative to less affluent contexts (Akadiri et al., 2022; Ridwan et al., 2024a). Although technological sophistication and regulatory standards may moderate some environmental impacts, sustained demographic expansion combined with consumption intensive lifestyles continues to place pressure on natural resources. This complexity reinforces the need to examine the long run association between population growth and ecological footprint within the specific structural conditions of the U.S. economy.

H4: Population growth increases the ecological footprint in the United States.

2.5 Artificial intelligence and ecological footprint

Artificial intelligence has emerged as a transformative technological innovation with significant potential to influence environmental outcomes. By enhancing the efficiency of energy management, production systems, and resource allocation, AI can contribute to decoupling economic growth from ecological degradation. Chen et al. (2022) demonstrated that AI applications in energy intensive industries improved operational efficiency and reduced emissions, while Liu et al. (2024) emphasized the role of AI driven technologies in supporting sustainable production practices and environmental monitoring. Evidence from Europe and China further confirms that AI adoption may promote ecological sustainability by reducing energy waste and facilitating clean technological advancement (Zhao et al., 2023; Dai, 2025). However, the environmental implications of AI are not uniformly positive. The expansion of AI systems requires substantial computational infrastructure, data processing capacity, and electricity consumption, which may increase overall energy demand. Moreover, productivity gains generated by AI can stimulate higher output and consumption levels, potentially offsetting efficiency improvements through scale or rebound effects. Therefore, while AI can function as an environmental optimizer, it may simultaneously intensify resource use depending on the broader economic and energy structure within which it operates. Recent investigations have examined the dynamics of AI in high income economies. Akther et al. (2024) observed that AI innovation contributed positively to ecosystem quality in the United States, whereas Su et al. (2021) reported similar sustainability enhancing effects for BRICS countries. Nevertheless, these findings do not eliminate the theoretical ambiguity surrounding AI's net environmental impact. The dual role of AI as both a technological efficiency enhancer and a potential driver of energy intensive expansion suggests that its overall effect on ecological footprint remains an empirical question. Despite the growing evidence, the literature remains limited and fragmented, particularly in examining AI alongside traditional drivers such as energy consumption, economic growth, industrialization, and population within the United States context.

H5: Artificial intelligence reduces the ecological footprint in the United States.

2.6 Literature gap

Environmental sustainability has been extensively examined, but there are still some critical gaps. The majority of previous studies uses CO_2 emissions as the only metrics of ecosystem, condition and ignored the wider ecological stress perspective. The ecological footprint that quantifies the total pressure on natural assets has not been well studied, particularly in industrialized countries such as the United States. Existing studies have also focused heavily on traditional drivers, such as growing GDP, energy use, industrialization, and population, while

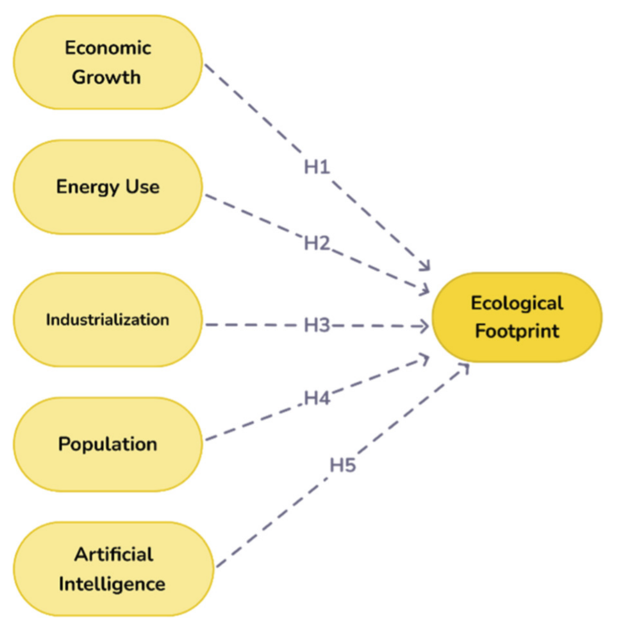

FIGURE 1
Hypothesis development of the study.

neglecting the role of emerging technologies. Despite its potential to improve efficiency, optimize resource use, and support cleaner production, artificial intelligence has received limited empirical attention in this context. In addition, many analyses have adopted narrow econometric approaches that lack robustness or ignore country-specific dynamics. These shortcomings underline the need for a comprehensive study that incorporates artificial intelligence into the environment–economy nexus using rigorous methods and focuses specifically on the United States. Figure 1 presents the hypotheses of this research.

3 Methodology

3.1 Data and variables

In this paper, annual time-series data from the United States between 1996 and 2024 were used. The selected period is determined by the availability of consistent AI-related patent data and ecological footprint statistics. Importantly, the time span captures both the early development phase of artificial intelligence and the more recent acceleration associated with advanced machine learning and digital transformation. The extended coverage up to 2024 allows the analysis to reflect contemporary structural shifts in technological innovation and environmental dynamics within the U.S. economy, thereby ensuring that both pre- and post-digital expansion phases are incorporated into the empirical framework.

The dependent variable used in this study is the ecological footprint (EF), which quantifies the total environmental pressure resulting from human activity. Unlike one-dimensional sustainability indicators such as CO_2 emissions, EF is a composite and multidimensional measure that integrates several forms of ecological demand, including energy consumption, cropland and forest use, built-up area, and other biocapacity components (Zafar et al., 2019). It thus provides a holistic representation of human demand on nature's regenerative capacity, capturing the balance between economic expansion and ecosystem resilience. While the ecological footprint is not without limitations—such as potential aggregation bias and incomplete coverage of certain ecological services—it remains one of the most comprehensive and internationally comparable indices for assessing overall environmental sustainability, particularly in studies adopting a macro- or cross-national perspective (e.g., Dogan et al., 2019; Hossain et al., 2026). Its multidimensional nature aligns with the objective of this research, which seeks to evaluate the combined effects of technological, financial, and demographic factors on environmental outcomes rather than focusing on a single source of pollution. Consequently, the EF serves as an appropriate and integrative proxy for measuring environmental sustainability, encapsulating the system-wide ecological consequences of economic and technological progress at the national level.

Artificial intelligence innovation is included as the primary explanatory variable of interest. AI represents advancement in technology that may lead to saving energy, to more environmentally friendly industrial work, and to more efficient utilization of resources (Liu et al., 2024). AI development is proxied by AI-related patent applications, which capture the intensity of inventive activity and technological progress within the AI domain. Although patent counts may not fully capture market adoption or commercialization, they are widely recognized as reliable indicators of the technological capability and innovative potential of an economy. Patents signal both the existence and direction of R&D efforts, embodying codified technical knowledge that forms the foundation for subsequent innovation and productivity growth. In the context of this study, AI-related patent activity therefore represents the innovation capacity and knowledge creation intensity underpinning AI evolution, rather than its downstream deployment across industries. This approach aligns with prior macro-level empirical studies linking technological innovation indicators to sustainability and economic transformation (e.g., Wang et al., 2025; Wen et al., 2025).

In addition to AI, four conventional drivers of ecological footprint were examined. Gross domestic product represents economic growth, which may increase environmental degradation through greater consumption and production (Ridwan et al., 2024b) but can also support ecological improvements through structural reforms. Energy consumption is considered due to its strong link to fossil fuel dependence and biodiversity loss (Nathaniel et al., 2021). Industrialization, measured as the share of industry in national output, captures the ecological implications of expanding production capacity (Aslam et al., 2024). Population growth is also included as demographic expansion intensifies resource use and ecological pressure (Khan et al., 2023). A summary of all variables, their measurements, expected signs, and data sources is presented in Table 1, which ensures transparency and supports the robustness of the empirical analysis. All factors were transformed into natural logarithms to

TABLE 1 Description and sources of the variables.

Variables	Measurements	Source
Ecological Footprint (EF)	Global hectares *per capita*	GFN (2024)
Artificial Intelligence (AI)	AI-related patents application	CSET (2024)
Gross Domestic Product (GDP)	GDP Per Capita (Constant US$)	WDI (2024)
Energy Consumption (ENU)	Energy use *per capita* (kg of oil equivalent)	WDI (2024)
Industrialization (INDUS)	Industry value-added (% of GDP)	WDI (2024)
Population (POP)	Total population	WDI (2024)

ensure comparability and facilitate an elasticity-based interpretation.

3.2 Theoretical framework and model specification

This study is grounded in two complementary theoretical perspectives that explain the relationship between socioeconomic development, technological progress, and environmental sustainability. First, Ecological Modernization Theory argues that advanced technologies and institutional improvements can reduce environmental degradation through efficiency enhancement, cleaner production processes, and improved resource management. Within this view, innovation serves as a mechanism that may decouple economic growth from ecological pressure by lowering environmental intensity. Second, this study builds upon the Stochastic Impacts by Regression on Population, Affluence, and Technology framework, which is an empirical extension of the IPAT identity originally proposed by Ehrlich and Holdren (1971). The IPAT model conceptualizes environmental impact as a multiplicative function of population, affluence, and technology. Although conceptually useful, the IPAT formulation is deterministic and does not permit statistical estimation. The STIRPAT framework transforms this identity into a stochastic regression model, allowing the elasticity of each determinant to be empirically tested.

Within the STIRPAT perspective, population growth increases ecological pressure through higher demand for energy, food, infrastructure, and natural resources. Affluence, proxied by GDP, reflects consumption intensity and scale effects that are often associated with environmental degradation. The technology component is theoretically ambiguous. Technological advancement may intensify environmental stress through industrial expansion and fossil fuel dependence, yet it may also mitigate ecological damage through innovation, efficiency gains, and structural transformation, consistent with ecological modernization arguments.

This study extends the STIRPAT model by explicitly incorporating artificial intelligence innovation as a contemporary technological factor. AI has the potential to reduce ecological footprint through optimization of resource use, intelligent energy management, and promotion of cleaner production systems. At the same time, AI development requires substantial computational infrastructure and electricity consumption, which may increase overall energy demand and production scale. Therefore, the environmental implications of AI are not predetermined and must be evaluated empirically. The modified STIRPAT formulation thus provides a coherent theoretical foundation for examining the combined effects of population, affluence, energy use, industrialization, and AI innovation on the ecological footprint of the United States. Equation 1 represents the IPAT identity:

$$I = P \times A \times T \tag{1}$$

The primary limitation of the IPAT framework is that it represents the influence of each factor on the environment as a fixed fraction. Dietz and Rosa (1994) revised the IPAT by adding stochastic effects with a combination of three variables to measure the ecological cost of population, affluence, and technological development. The new formulation is given below in Equation 2.

$$I_{it} = CP_{it}^{\omega_1} A_{it}^{\omega_2} T_{it}^{\omega_3} \varepsilon_{it} \tag{2}$$

The constant factor in the STIRPAT model is C, and the random error factor is ε. Conversely, P, A, and T are represented by ω_1, ω_2 and ω_3.The logarithmic form of the model is as follows in Equation 3:

$$ln\, I_{it} = C + \omega_1\, ln\, P_{it} + \omega_2\, ln\, A_{it} + \omega_3\, ln\, T_{it} + \varepsilon_{it} \tag{3}$$

Furthermore, the mathematical framework established in Equation 4 is an experimental version based on currently available literature.

$$EF = f\,(GDP_{it}, AI_{it}, ENU_{it}, INDUS_{it}, POP_{it}) \tag{4}$$

Details of the variables in Equation 4 are presents in Table 1. With values ranging from 1 to N, the variables 'i' and 't' stand for the countries and timeframe, respectively, of the study. This study applied the natural logarithm of Equation 5, which could be presented as follows, to solve the problem of heteroscedasticity and improve the data quality.

$$\begin{aligned} ln\, EF_{it} = \beta_0 + \beta_1 lnGDP_{it} + \beta_2 lnAI_{it} + \beta_3 lnENU_{it} + \beta_4 lnINDUS_{it} \\ + \beta_5 lnPOP_{it} + \varepsilon_{it} \end{aligned} \tag{5}$$

Where β_0 = the individual intercept expression, $\beta_1, \beta_2, \beta_3, \beta_4$, β_5 = the elasticity of the exogenous variables, and ln EF, ln GDP, ln AI, ln ENU, and ln POP are the logarithmic formations of EF, GDP, AI innovation, energy consumption, industrialization, and population size, respectively.

3.3 Empirical strategy

Empirical evaluation is a multistep plan with strong and reliable outcomes. To test stationarity, the time-series characteristics of all variables are taken into account in terms of the Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and Dickey-Fuller Generalized Least squares (DF-GLS) tests. These complementary approaches help to account for serial correlation, heteroscedasticity, and deterministic components, thereby ensuring that the data are appropriate for cointegration analysis. Furthermore, this paper incorporated the ARDL bounds-testing method developed by Pesaran and Shin (1995). This approach is particularly appropriate for small samples and can be used to concurrently estimate the I (0) and I (1) variables. It also allows for some flexibility in the lag structures and provides the opportunity to estimate the long- and short-run dynamics. To confirm the presence of cointegration between EF, AI, GDP expansion, energy usage, industrialization, and population, a bounds test was used. After cointegration, a correction model (ECM) is calculated to represent the pace of the long-run equilibrium between the short-run deviations. The ARDL bound test is shown in Equation 6.

$$\begin{aligned}\Delta LEF_t = {} & \delta_0 + \delta_1 LEF_{t-1} + \delta_2 LGDP_{t-1} + \delta_3 LAI_{t-1} + \delta_4 LENU_{t-1} \\ & + \delta_5 LINDUS_{t-1} + \delta_6 LPOP_{t-1} + \sum_{i=1}^{m} \gamma_1 \Delta LEF_{t-i} \\ & + \sum_{i=1}^{m} \gamma_2 \Delta LGDP_{t-i} + \sum_{i=1}^{m} \gamma_3 \Delta LAI_{t-i} + \sum_{i=1}^{m} \gamma_4 \Delta LENU_{t-i} \\ & + \sum_{i=1}^{m} \gamma_5 \Delta LINDUS_{t-i} + \sum_{i=1}^{m} \gamma_6 \Delta LPOP_{t-i} + \varepsilon_t \end{aligned} \tag{6}$$

The ECM formulation is given in Equation 7:

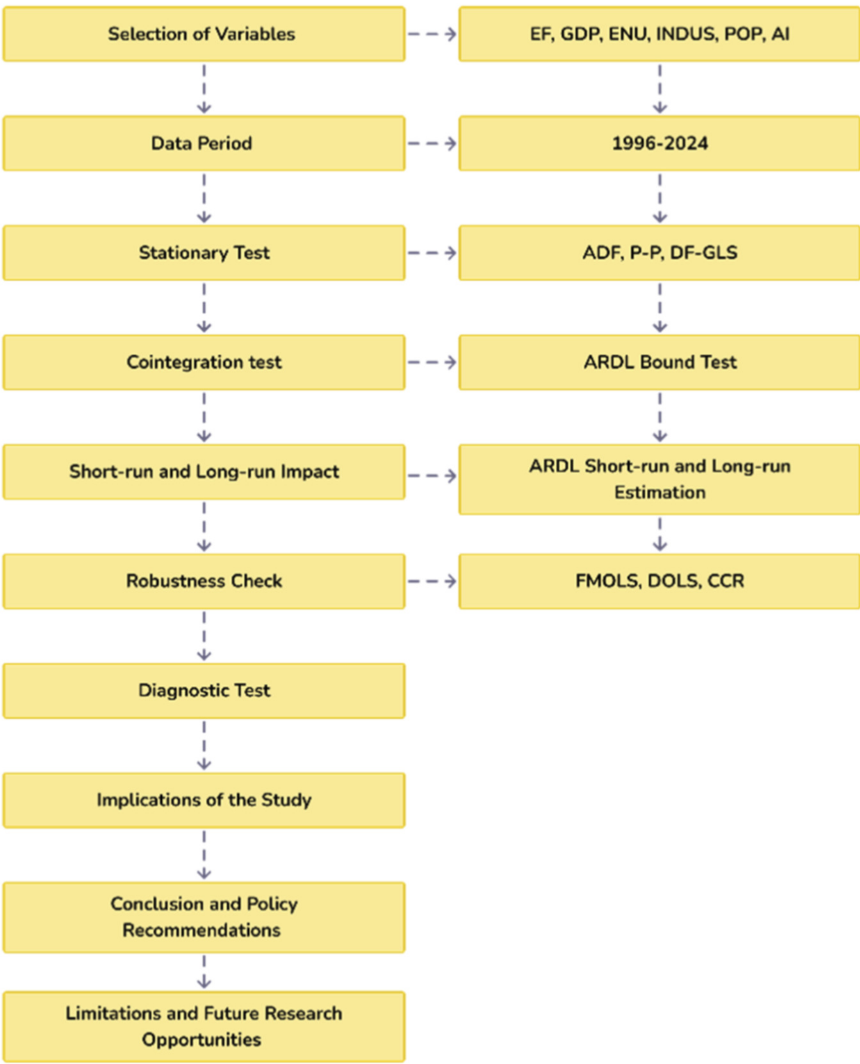

FIGURE 2
Flowchart of estimation methods.

TABLE 2 Summary statistics.

Variables	Mean	Median	Maximum	Minimum	Standard deviation	Skewness	Kurtosis	Jarque bera	Probability
LEF	3.45	3.4	3.9	3.05	0.25	0.18	2.45	1.12	0.57
LAI	1.27	1.25	1.45	1.1	0.11	0.22	2.6	1.25	0.54
LGDP	10.68	10.65	11.05	10.2	0.22	0.05	2.05	0.98	0.61
LENU	7.92	7.88	8.2	7.6	0.15	0.14	2.35	1.05	0.59
LINDUS	25.34	25.1	27	23.8	0.82	0.30	2.55	1.34	0.51
LPOP	8.65	8.64	8.72	8.58	0.04	−0.10	2.1	0.88	0.64

$$\begin{aligned}\Delta LEF_t &= \delta_0 + \delta_1 LEF_{t-1} + \delta_2 LGDP_{t-1} + \delta_3 LAI_{t-1} + \delta_4 LENU_{t-1} \\ &+ \delta_5 LINDUS_{t-1} + \delta_6 LPOP_{t-1} + \sum_{i=1}^{m} \gamma_1 \Delta LEF_{t-i} \\ &+ \sum_{i=1}^{m} \gamma_2 \Delta LGDP_{t-i} + \sum_{i=1}^{m} \gamma_3 \Delta LAI_{t-i} + \sum_{i=1}^{m} \gamma_4 \Delta LENU_{t-i} \\ &+ \sum_{i=1}^{m} \gamma_5 \Delta LINDUS_{t-i} + \sum_{i=1}^{m} \gamma_6 \Delta LPOP_{t-i} + ¥ECM_{t-1} + \varepsilon_t \end{aligned} \tag{7}$$

Moreover, it is performed using alternative cointegration estimators, that is, FMOLS, DOLS, and CCR. To evaluate the consistency of the results, these estimators help overcome possible endogeneity and serial correlation issues. Lastly, diagnostic and stability assessments were conducted to assure the consistency of the estimated models. The Breusch-Godfrey Lagrange Multiplier test was used to assess autocorrelation, the Breusch-Pagan-Godfrey test was adopted to evaluate heteroscedasticity, and the Jarque-Bera test was employed to ensure the normality of the residuals. CUSUM and CUSUMSQ tests verified the validity of the models. The combination of this empirical approach offers a strict methodology to study the implication of AI and other drivers on the EF in the United States. Figure 2 shows a flowchart of the estimation approach.

4 Results and discussion

Table 2 represents descriptive statistics for the study factors. The EF was 3.45 with a range of 3.05–3.90 and shows that environmental pressure was not too stationary or varied within the sample period. AI has a mean of 1.27 and a low standard deviation, which indicates that technological innovation grows consistently but lacks high volatility. The average GDP is 10.68, which is in line with the stable long-term performance of the U.S. economy. ENU is averaged at 7.92, which indicates a high energy dependence in the country, whereas INDUS is averaged at 25.34 percent of GDP, which still indicates the significance of industrial activity in the economy. The POP does not change significantly, with an average of 8.65, and the difference between observations is insignificant. Altogether, these findings indicate that although GDP, ENU, INDUS, and POP tend to contribute to increasing pressure on EF, AI has an alternative pattern of behavior that promises to reduce ecological stress.

ADF, PP, and DF-GLS tests were used to determine the stationarity of the factors, as shown in Table 3. The results show a mixed pattern of integration, with certain variables remaining at that level and some attaining stationarity after the initial stage of differentiation. In particular, both EF and ENU are observed to be I (0), indicating that both series are stationary without differencing. On the other hand, GDP, AI, INDUS, and POP are non-stationary at the levels but stationary in the first differences; that is, they are integrating order I (1). Because none of the variables are included in I (2), the use of the ARDL bounds testing technique is adequate in that both I (0) and I (1) variables can be included in the same estimation equation.

Table 4 provides the results of the ARDL bounds-testing procedure. The calculated F-statistic was 6.14 and exceeds the critical upper values at the 1%, 2.5%, 5%, and 10% levels. This provides a good indication of the long-run cointegrating relationship between EF, GDP, AI, ENU, INDUS, and POP. In other words, despite the variables being integrated in mixed orders, they move together in the long term, thereby confirming the suitability of the ARDL framework. Confirmation of cointegration has important implications for the analysis. This suggests that the selected factors jointly determine the long-run dynamics of EF in the United States of America. While short-run fluctuations may occur because of shocks or cyclical changes, the system consistently returns to a stable equilibrium. This confirms the subsequent estimation of both the short- and long-run ARDL models and the ECM to measure the adjustment speed.

The ARDL findings indicate that GDP has a positive and statistically significant influence on ecological footprint in both the long and short run. In the long run, the coefficient is 0.354 ($p < 0.01$), while in the short run it is 0.182 ($p < 0.05$), suggesting that economic expansion is associated with increased environmental pressure in the United States (Tables 5, 6). This result reflects the close connection between rising GDP, industrial production, urbanization, and energy consumption, all of which contribute to greater ecological demand. Although economic growth enhances income, infrastructure, and overall welfare, it remains closely linked to fossil fuel dependence, energy intensive consumption, and large scale industrial operations. These structural characteristics reinforce the environmental cost of growth, indicating that economic expansion continues to exert pressure on sustainability in the U.S. context. This finding aligns with the scale effect predicted by the STIRPAT framework, where affluence increases environmental pressure through higher consumption intensity. Similar evidence

TABLE 3 Unit root test.

Variable	ADF level	ADF 1st diff	PP level	PP 1st diff	DF-GLS level	DF-GLS 1st diff	Order of integration
LEF	−2.42*	-	−2.55*	-	−2.10*	-	I (0)
LGDP	−1.84	−4.88***	−1.90	−4.95***	−1.42	−4.47***	I (1)
LAI	−2.05	−5.15***	−2.14	−5.20***	−1.58	−4.78***	I (1)
LENU	−2.65**	-	−2.72**	-	−2.30**	-	I (0)
LINDUS	−2.12	−5.27***	−2.18	−5.33***	−1.60	−4.81***	I (1)
LPOP	−1.75	−4.63***	−1.82	−4.71***	−1.33	−4.36***	I (1)

Note: $p < 0.01$*, $p < 0.05$**, $p < 0.01$***

TABLE 4 ARDL bound test.

Test statistic	Value	Significance	Critical values
F-statistic	6.14	10%	2.26 (I0), 3.35 (I1)
		5%	2.62 (I0), 3.79 (I1)
		2.5%	2.96 (I0), 4.18 (I1)
		1%	3.41 (I0), 4.68 (I1)

TABLE 5 ARDL long-run estimation.

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LGDP	0.354	0.082	4.32	0.000
LENU	0.412	0.099	4.16	0.000
LINDUS	0.227	0.092	2.47	0.020
LPOP	0.318	0.123	2.58	0.015
LAI	−0.139	0.054	−2.57	0.016
C	1.047	0.408	2.57	0.017

TABLE 6 ARDL short-run estimation.

Variable	Coefficient	Std. Error	t-Statistic	p-Value
ΔLGDP	0.182	0.071	2.56	0.017
ΔLENU	0.196	0.082	2.39	0.022
ΔLINDUS	0.115	0.058	1.98	0.056
ΔLPOP	0.152	0.071	2.14	0.041
ΔLAI	−0.072	0.031	−2.32	0.025
ECT (-1)	−0.562	0.112	−5.02	0.000

has been reported for Bangladesh, China, and BRICS economies (Rahman et al., 2023; Ridwan et al., 2024b; Ahmad et al., 2024). Comparable results were also identified by Nketiah et al. (2024) and Saud et al. (2019), while Mehmood et al. (2023) suggested that advanced economies may eventually achieve partial decoupling under specific structural conditions. However, the present evidence indicates that the United States continues to experience a growth and environment trade off. Therefore, the results support the hypothesis that economic expansion is positively associated with ecological footprint and highlight the importance of integrating green innovation and sustainable energy strategies into long term growth planning.

The ARDL estimates indicate that energy consumption has a positive and statistically significant effect on ecological footprint in both the long and short run. The long run coefficient is 0.412 ($p < 0.01$), while the short run coefficient is 0.196 ($p < 0.05$), suggesting that higher energy use is associated with increased environmental pressure in the United States (Tables 5, 6). This outcome reflects the continued reliance of the U.S. economy on fossil fuel based energy sources, which remain a substantial share of the national energy mix despite recent growth in renewable energy capacity. Increased energy consumption across industrial production, transportation, and households contributes to higher emissions and ecological demand. From the STIRPAT perspective, this result aligns with the scale effect, where greater energy use intensifies environmental pressure, particularly in economies with significant carbon intensive infrastructure. Although renewable energy expansion has the potential to mitigate ecological degradation, total energy consumption continues to exert upward pressure on ecological footprint in the U.S. context. These findings are consistent with prior evidence showing that energy consumption increases ecological footprint in Vietnam, the United Kingdom, and Next Eleven economies (Nathaniel et al., 2021; Eweade et al., 2024; Javeed et al., 2023). In contrast, Rahman et al. (2023) demonstrated that renewable energy adoption can reduce ecological footprint in large economies, suggesting that structural transformation in energy composition may alter the long run relationship. Overall, the results support the hypothesis that energy consumption is positively associated with ecological footprint in the United States and emphasize the importance of accelerating the transition toward cleaner energy sources to moderate long term environmental pressure.

According to the ARDL estimates reported in Tables 5, 6, industrialization exerts a statistically significant positive impact on ecological footprint in both the long and short run. In the long run, the coefficient is 0.227 ($p < 0.05$), while in the short run it is 0.115 ($p < 0.10$), indicating that an increase in industrial activity is associated with greater ecological pressure in the

United States. This result reflects the environmental cost of large scale production and manufacturing processes, which are typically energy intensive and often reliant on fossil fuel based inputs. Expansion of industrial activity contributes to higher resource extraction, emissions, and waste generation, thereby increasing overall ecological demand. From the STIRPAT perspective, this outcome is consistent with the scale effect, whereby expansion of productive capacity intensifies environmental stress unless offset by significant technological improvements. Although industry remains a fundamental engine of economic growth, its environmental consequences depend on the balance between structural expansion and technological efficiency. In the U.S. context, the persistence of energy intensive industrial segments may continue to exert upward pressure on ecological footprint. These findings are consistent with previous research that documented adverse environmental effects of industrialization in East Asia (Aslam et al., 2024) and in BRICS, OECD, and MENA economies (Kongbuamai et al., 2021; Nasrollahi et al., 2020). Patel and Mehta (2023) suggested that industrialization may eventually reduce emissions under structural transformation and cleaner production systems. However, the present evidence indicates that industrial expansion in the United States remains positively associated with ecological footprint. Therefore, the results support the hypothesis that industrialization contributes to environmental pressure and underscore the importance of promoting cleaner production technologies within industrial policy frameworks.

The ARDL results indicate that population growth has a positive and statistically significant effect on ecological footprint in both the long and short run. In the long run, the coefficient is 0.318 ($p < 0.05$), while in the short run it is 0.152 ($p < 0.05$), suggesting that demographic expansion is associated with increased ecological pressure in the United States. This finding reflects the contribution of population growth to rising demand for food, housing, transportation, and energy, which collectively intensify environmental stress. Even modest population increases can elevate consumption needs in the short run, while sustained demographic expansion may amplify ecological pressure through urbanization and higher industrial demand. Within the STIRPAT framework, this outcome is consistent with the scale effect, where population growth increases environmental impact through expanded resource use. The U.S. context is particularly relevant given its high *per capita* consumption levels, which magnify the environmental consequences of demographic expansion (Caglar et al., 2022). These findings align with previous evidence from South Asia and Africa, where population growth significantly intensified ecological degradation (Khan et al., 2023; Nathaniel et al., 2021). Although some advanced economies may partially offset demographic pressures through technological progress (Abbasi et al., 2022), the present results suggest that population growth in the United States remains positively associated with ecological footprint. Therefore, the findings support the hypothesis that demographic expansion contributes to environmental pressure and highlight the importance of integrating sustainable urban planning and resource efficiency into long term development strategies.

The ARDL results indicate that artificial intelligence innovation has a negative and statistically significant association with ecological footprint in both the long and short run. The long run coefficient is −0.139 ($p < 0.05$), while the short run coefficient is −0.072 ($p < 0.05$), suggesting that greater AI innovation is associated with reduced ecological pressure in the United States. This finding implies that AI development may contribute to a relative decoupling between economic expansion and environmental degradation. The integration of AI technologies can enhance energy efficiency, optimize industrial operations, and strengthen environmental monitoring systems, thereby lowering overall ecological demand. In the short run, efficiency gains may arise from improved resource allocation, whereas in the long run, deeper integration of AI may generate structural adjustments in production processes and resource management practices. Interpreted through the lens of Ecological Modernization Theory, this result suggests that advanced technological innovation can function as a mechanism for reducing environmental intensity within a high income economy. Within the STIRPAT framework, the negative coefficient indicates that the efficiency channel of technological progress outweighs potential scale and energy demand effects associated with AI infrastructure during the study period. These findings are consistent with prior evidence indicating that AI innovation can mitigate ecological stress in the United States and G-7 economies (Zafar et al., 2019; Raihan et al., 2024b), as well as in China and Europe, where AI has supported transitions toward cleaner production and sustainable energy systems (Zhao et al., 2023; Dai, 2025). Overall, the results support the hypothesis that AI innovation is negatively associated with ecological footprint in the United States and highlight the importance of technology driven sustainability strategies in advanced economies.

Table 7 presents the results of the Wald joint parameter stability test for the long run ARDL model. The null hypothesis states that the estimated long run slope coefficients of GDP, AI, ENU, INDUS, and POP are jointly stable over the sample period. The Chi square statistic of 8.472 with 5 degrees of freedom yields a probability value of 0.132, which exceeds the conventional 5 percent significance level. Therefore, the null hypothesis cannot be rejected. This result indicates that the long run coefficients remain structurally stable throughout the study period. The stability of parameters suggests that the estimated relationships between ecological footprint and its determinants are not driven by structural breaks or instability, thereby reinforcing the robustness and reliability of the ARDL long run estimates.

This study performed additional estimates using the FMOLS, DOLS, and CCR methods to verify ARDL results. These data are presented in Tables 8–10. The trends and meaningfulness of the coefficients are in line with the ARDL estimated values for each of the three approaches, which justifies the original findings. Its role as a factor of EF in the United States is further evidenced by the positive and very high coefficients of GDP in FMOLS (0.361, $p < 0.01$), DOLS (0.348, $p < 0.01$), and CCR (0.357, $p < 0.01$), as presented in Tables 8–10. Similarly, ENU has a positive and significant influence across all estimators, with a coefficient between 0.408 and 0.421, which once again highlights the devastation of the environment by fossil fuel reliance. The positive effect of INDUS is also maintained, with an estimation of 0.219–0.233 ($p < 0.05$), showing that industrial activity results in increased EF. The POP effect did not change, and the coefficients were found to be in the 0.309 to 0.325 range, all

TABLE 7 Wald joint stability test.

Test	Chi-square statistic	Degrees of freedom	Probability
Joint stability of long-run slopes	8.472	5	0.132

TABLE 8 Fully modified OLS (FMOLS) results.

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LGDP	0.361	0.085	4.25	0.000
LENU	0.408	0.102	4.00	0.000
LINDUS	0.233	0.095	2.45	0.021
LPOP	0.325	0.126	2.58	0.015
LAI	−0.144	0.056	−2.57	0.016

TABLE 9 Dynamic OLS (DOLS) results.

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LGDP	0.348	0.081	4.29	0.000
LENU	0.421	0.098	4.29	0.000
LINDUS	0.219	0.090	2.43	0.022
LPOP	0.309	0.121	2.55	0.016
LAI	−0.137	0.052	−2.63	0.014

TABLE 10 Canonical cointegrating regression (CCR) results.

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LGDP	0.357	0.083	4.30	0.000
LENU	0.415	0.100	4.15	0.000
LINDUS	0.226	0.092	2.46	0.020
LPOP	0.317	0.124	2.56	0.016
LAI	−0.141	0.055	−2.56	0.016

significant at the 5% level, showing the implication of demographic pressure on ecological resources. More significantly, AI had a significant negative effect on FMOLS (−0.144, $p < 0.05$), DOLS (−0.137, $p < 0.05$), and CCR (−0.141, $p < 0.05$), confirming its cause-reducing effect on EF. In summary, the robustness checks establish that the ARDL findings are not dependent on estimation methods, which gives more confidence in the conclusion that traditional drivers positively affect the increase in EF and that the innovation of AI can enhance ecosystem health in the United States.

Several diagnostic and stability assessments were performed to guarantee that the computed ARDL model was reliable. Table 11 presents the study outcomes. With a p-value of 0.41 and a statistic of 1.82, the Breusch-Godfrey LM test concludes that the residuals do not exhibit serial correlation. The Jarque-Bera statistic (1.37; $p = 0.50$) further confirms the normal distribution of the residuals. Taken together, these results indicate that the main assumptions of the ARDL framework were met, which means that the research findings are solid. CUSUM and CUSUMSQ tests were used to expand the reliability of the model. Figure 3 ensured the strong foundation of the model; both plots remain within the 5% critical boundaries throughout the sample time. This means that there is no impact from structural fractures or fragility on the model and that the calculated parameters remain constant throughout time.

5 Implications of the study

5.1 Theoretical implications

This analysis offers significant contributions to the theoretical understanding of the economy–environment nexus by extending the traditional STIRPAT framework. While the STIRPAT model has long emphasized population, affluence, and technology as the main determinants of environmental impact, this research explicitly incorporates artificial intelligence as a novel technological factor. The empirical outcomes demonstrate that AI exerts a mitigating influence on EF, in contrast to conventional drivers, such as GDP, ENU, INDUS, and POP, which increase ecological stress. This theoretical innovation strengthens the argument that technological progress cannot be viewed solely as a source of environmental pressure but also as a pathway to ecological resilience. Furthermore, by applying novel econometric methods such as ARDL, FMOLS, DOLS, and CCR, this study contributes to methodological development in environmental economics, demonstrating that robust empirical strategies can enhance theoretical validation. Overall, the findings advance the sustainability theory by positioning AI as a transformative element in explaining environmental outcomes, thereby enriching the existing literature and opening new avenues for research on digital technologies within ecological frameworks.

5.2 Managerial implications

These results also provide crucial practical suggestions for businesses, industries, and energy planners. The positive role of GDP, ENU, INDUS, and POP in increasing EF underscores that growth and industrial expansion remain resource-intensive in the U.S. context. At the same time, the negative effect of AI highlights its potential as a managerial tool to optimize resource allocation, reduce garbage, and enhance production procedures. Industries that adopt AI-driven technologies in energy management, logistics, and manufacturing can significantly reduce their ecological footprints while maintaining competitiveness. For energy planners, evidence suggests that integrating AI into renewable energy systems can accelerate the shift from fossil fuels, ensuring a cleaner and more efficient energy supply. For urban managers, AI offers solutions for

TABLE 11 Diagonstic test.

Test	Statistic	p-Value	Decision
Breusch-godfrey LM test	1.82	0.41	No serial correlation
Breusch-pagan-godfrey test	6.24	0.28	No heteroscedasticity
Jarque-bera test	1.37	0.50	Residuals are normally distributed

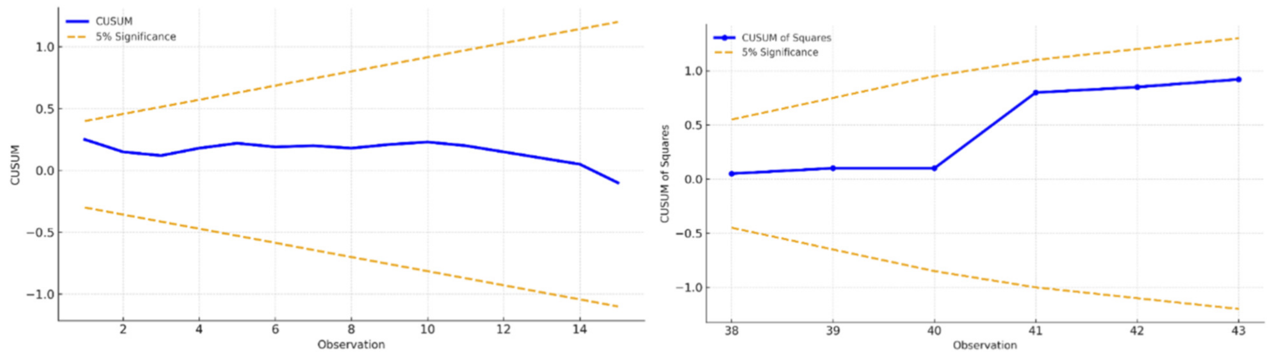

FIGURE 3
CUSUM and CUSUMSQ test.

traffic optimization, waste management, and sustainable urban planning, thereby reducing the environmental burden of growing populations. These insights underscore that sustainability is not only a policy challenge, but also a strategic management issue, where technological innovation can serve as a critical driver of ecological performance at the firm and sectoral levels.

6 Conclusion and policy recommendations

6.1 Conclusion

Using yearly data from 1996 to 2024, this study tested the hypothesis that the EF in the US is affected by GDP growth, energy consumption, industrialization, population growth, and AI. The results indicate that the factors are consistently interconnected in the long run when the ARDL bounds testing strategy is applied to the FMOLS, DOLS, and CCR estimation instruments. The results indicate that GDP, ENU, INDUS, and POP significantly boost EF in the short and long term. This is because business activities, energy demand, manufacturing expansion, and population growth all require resources. On the other hand, advancements in AI have a noticeable and detrimental impact, which indicates its potential as a technological solution to alleviate environmental stress. The robustness of these results was confirmed through multiple estimators and supported by diagnostic and stability tests that verified the reliability of the model. Collectively, the findings advance theoretical understanding by extending the STIRPAT framework to incorporate AI as a mitigating factor in the growth–environment nexus. They also offer practical guidance for industries and energy managers, while providing strong policy insights for integrating AI into national sustainability strategies.

6.2 Policy recommendations

To curb the current increase in EF and maintain growth, policymakers must integrate AI as a key component of their sustainability initiatives. At the energy level, AI-based forecasting and grid optimization are required for utilities to accelerate the process of renewable integration and minimize the use of fossil fuels. Federal tax credits and green financing should incentivize companies to implement AI-based emissions monitoring, predictive maintenance, and resource-efficient production systems. Urban policy should also focus on AI-enhanced traffic regulation and waste and water recycling and management, particularly in densely populated cities where the population strains the environmental conditions. An AI-Climate Innovation Fund should be created to amplify research and the commercialization of AI solutions to capture carbon, precision agriculture, and resilient infrastructure. Finally, an act of parliament, the AI for Sustainability Act, must bring federal efforts to SDGs 7, 9, 11, and 13 and institutionalize AI in climate, energy, and industrial policy. Combined, these actions make AI a strategic resource that can help decouple growth and environmental degradation, and place the U.S. at the forefront of the digital-green transition.

6.3 Limitations and future research opportunities

Although this study provides important insights into the interaction between economic growth, energy consumption, industrialization, population, and artificial intelligence (AI) innovation in shaping the ecological footprint of the United States, several limitations should be acknowledged. First, the analysis focuses on a single high-income economy, which may constrain the generalizability of the findings to emerging or structurally distinct contexts. A second limitation relates to the measurement of AI innovation. In this study, AI innovation is proxied using national-level patent data, which effectively capture inventive activity but may not fully reflect the breadth, quality, or commercialization intensity of AI development. Patents represent upstream innovation capacity but do not capture differences in technological maturity, R&D expenditure, or adoption rates across sectors. Consequently, the index may understate the extent to which AI is actively deployed in energy-intensive or sustainability-oriented industries such as manufacturing or renewables.

Third, while the ecological footprint provides a comprehensive measure of aggregate environmental stress, it does not capture component-level ecological effects such as carbon intensity, water resource depletion, or biodiversity loss. Moreover, although the STIRPAT framework enables estimation of the elasticities of demographic, economic, and technological drivers, it does not explicitly model potential nonlinearities, feedback loops, or stage-specific dynamics in technological diffusion and environmental response. Future research may therefore extend the current framework in several directions: (i) employing sector-level or firm-level AI adoption indicators to discern heterogeneous environmental effects across industries; (ii) exploring cross-country comparisons to account for institutional, technological, and policy differences; (iii) incorporating disaggregated environmental outcome metrics to evaluate how distinct ecological dimensions respond to AI-driven innovation under varying structural, regulatory, and technological conditions; and (iv) expanding the concept of technological progress to include other aspects of AI innovation beyond patents, such as R&D investment, AI infrastructure, and market deployment indicators.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

MR: Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Software, Visualization, Writing – original draft, Writing – review and editing. JK: Conceptualization, Investigation, Methodology, Project administration, Writing – original draft, Writing – review and editing. CL: Resources, Software, Writing – original draft, Writing – review and editing. W-KM: Formal Analysis, Funding acquisition, Supervision, Writing – review and editing.

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